MIT-WPU | Faculty of Engineering and Technology

School of Computer Science and Engineering
Department of Computer Engineering and Technology

S. Y. B. Tech CSE **AY 2026-27** **Semester: 3**

Computer Networks Lab (CSE40110)

Assignment No.: 7

Title: DHCP, DNS, Web Server and Email configuration in cisco packet tracer.

Aim: DHCP, DNS and Web Server configuration (Use: Packet Tracer/GNS). Configure network using Dynamic Host Configuration Protocol (DHCP), DNA and Web Server Use Ping utility to test connectivity.

Objectives: To learn the DHCP, DNS and Web Server installation and understand their practical usage.

Theory: (Handwritten & Scanned)

- Student's should write One-page theory report (Handwritten & Scanned)
  - Dynamic Host Configuration Protocol (DHCP) (Functionality and need)
  - DNS (Domain Name System resource records, iterative and recursive lookups)
  - SMTP, POP3 and IMAP (Functionality and port numbers)

- **Design Screenshots:**

  Paste design screen shots

- **Output Screenshots:**

  Student's should write/paste Code here (Screenshots/Snaps are not accepted)

- **Students Observation:**

  Write your observations on the mentioned objective

FAQ (Handwritten & Scanned)

1. What are the different DHCP messages involved in the dynamic IP address allocation process?
2. Explain the DORA process used by DHCP for dynamic IP address allocation.
3. What are the different types of Resource Records (RRs) used in DNS? Explain their purpose.
4. Compare SMTP, POP3, and IMAP based on their functions and methods of handling email messages.

Computer Networks Lab (CSE40110) - Assignment No.: 7 (Continued)

Name: _____ Roll No.: _____ Batch: _____

Theory:

1. Dynamic Host Configuration Protocol (DHCP)

DHCP is an application layer protocol that gives IP configuration to hosts automatically. A DHCP server hands out an IP address, subnet mask, default gateway, DNS server address and a lease time. It uses UDP port 67 on the server and UDP port 68 on the client. Without DHCP, an administrator must set every host by hand. That is slow, and one duplicate or mistyped address breaks connectivity. DHCP removes this work, avoids address conflicts, reuses addresses when hosts leave (through lease expiry) and lets a change such as a new gateway be made once on the server. The exchange is Discover, Offer, Request, Acknowledge (DORA).

2. DNS (Domain Name System)

DNS maps human readable names such as www.mitwpu.edu to IP addresses. It is a distributed, hierarchical database (root, TLD, authoritative servers) that uses UDP/TCP port 53. Data is stored as resource records (RR): **A** (name to IPv4), **AAAA** (name to IPv6), **CNAME** (alias to canonical name), **NS** (authoritative name server of a zone), **MX** (mail server of a domain), **PTR** (IP to name), **SOA** (start of authority, zone details) and **TXT** (free text, e.g. SPF).

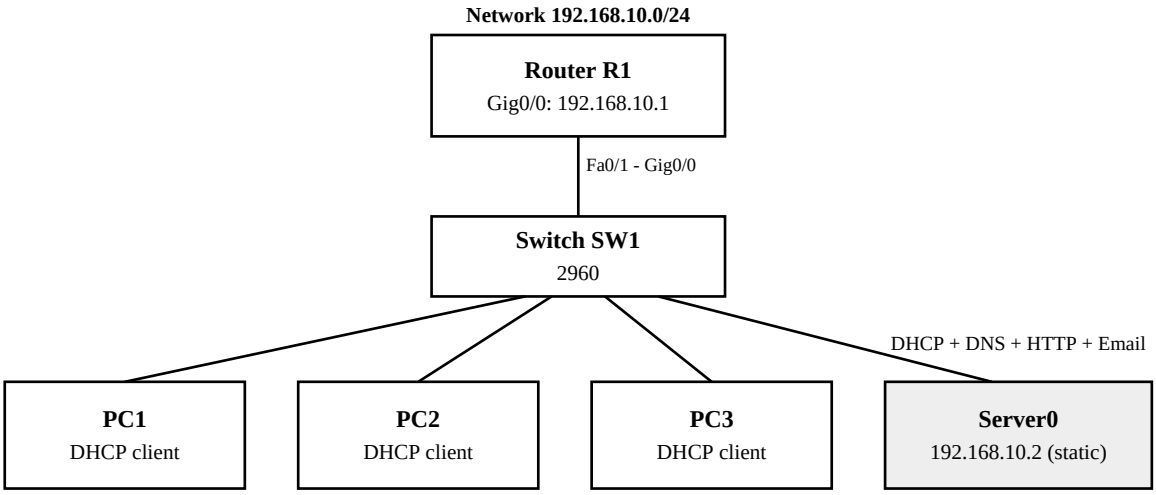
Recursive lookup: the client asks its resolver for the final answer. The resolver does all the work and returns either the IP address or an error. **Iterative lookup:** a server never chases the answer. It replies with the best answer it has or a referral to another server. For www.mitwpu.edu the resolver asks a root server (referral to .edu servers), then a .edu server (referral to the mitwpu.edu name server), then that server, which returns the A record. Usually the client-to-resolver step is recursive and the resolver-to-server steps are iterative.

3. SMTP, POP3 and IMAP

SMTP (Simple Mail Transfer Protocol) pushes mail from a client to a mail server and between mail servers. It is a push protocol. Ports: 25 (server to server), 587 (client submission with STARTTLS), 465 (SMTP over SSL). **POP3** (Post Office Protocol v3) pulls mail from the server to one device, and by default deletes it from the server after download. Ports: 110 (plain), 995 (SSL/TLS). **IMAP** (Internet Message Access Protocol) lets the client read and manage mail on the server, so folders, read flags and messages stay in sync across devices. Ports: 143 (plain), 993 (SSL/TLS).

Design Screenshots:

Topology built in Cisco Packet Tracer: one router, one switch, three PCs set to DHCP and one server (Server0) that runs the DHCP, DNS, HTTP and Email services.



Device	Interface	IP Address	Subnet Mask	Gateway
R1	Gig0/0	192.168.10.1	255.255.255.0	N/A
Server0	Fa0	192.168.10.2 (static)	255.255.255.0	192.168.10.1
PC1, PC2, PC3	Fa0	DHCP (192.168.10.10 onwards)	255.255.255.0	192.168.10.1

Server0 service settings

Service	Setting
DHCP (ON)	Pool name: LAN_POOL | Default gateway: 192.168.10.1 | DNS server: 192.168.10.2 | Start IP: 192.168.10.10 | Subnet mask: 255.255.255.0 | Max users: 50 | click Save
DNS (ON)	www.mitwpu.edu A 192.168.10.2 | mail.mitwpu.edu A 192.168.10.2 | web.mitwpu.edu CNAME www.mitwpu.edu
HTTP (ON)	Default index.html edited to show: Welcome to MIT-WPU CN Lab
EMAIL (ON)	Domain name: mitwpu.edu | Users: user1 / pass 1234, user2 / pass 1234 (SMTP and POP3 enabled)

Email client on PC1 / PC2 (Desktop > Email > Configure Mail)

Field	PC1	PC2
Your Name	User One	User Two
Email Address	user1@mitwpu.edu	user2@mitwpu.edu
Incoming / Outgoing Mail Server	mail.mitwpu.edu	mail.mitwpu.edu
User Name / Password	user1 / 1234	user2 / 1234

Output Screenshots:

A. Router R1 configuration (CLI)

```
Router>enable
Router#configure terminal
Router(config)#hostname R1
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
R1(config)#end
R1#write memory
Building configuration...
[OK]
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.10.1    YES manual up                    up
```

B. Server0 static IP (Desktop > IP Configuration)

```
IPv4 Address    : 192.168.10.2
Subnet Mask     : 255.255.255.0
Default Gateway : 192.168.10.1
DNS Server      : 192.168.10.2
```

C. PC1: IP Configuration set to DHCP, then Command Prompt

```
C:\>ipconfig /renew

   IP Address......................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.10.1
   DNS Server......................: 192.168.10.2

C:\>ipconfig /all

FastEthernet0 Connection:(default port)
   Physical Address................: 0001.4321.A1B2
   IPv4 Address....................: 192.168.10.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: 192.168.10.1
   DHCP Servers....................: 192.168.10.2
   DNS Servers.....................: 192.168.10.2
```

D. Connectivity test with ping

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

E. DNS test

```
C:\>nslookup www.mitwpu.edu
Server: [192.168.10.2]
Address: 192.168.10.2

Non-authoritative answer:
Name:    www.mitwpu.edu
Address: 192.168.10.2

C:\>ping www.mitwpu.edu

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

F. Web server test (Desktop > Web Browser, URL: http://www.mitwpu.edu)

```
Page loaded: "Welcome to MIT-WPU CN Lab"
```

G. Ping between DHCP clients (PC1 to PC2)

```
C:\>ping 192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

H. Email test (PC1 sends to PC2)

```
PC1 > Email > Compose
   To      : user2@mitwpu.edu
   Subject : CN Lab Test
   Body    : Hello from PC1
   Click Send        -> "Mail Sent Successfully"
PC2 > Email > Receive -> 1 new mail from user1@mitwpu.edu received
```

Students Observation:

- PC1, PC2 and PC3 received IP addresses automatically from Server0 (192.168.10.10, .11, .12) in sequence, along with mask, gateway and DNS server. No manual entry was needed on the PCs.
- The server and router interface use static addresses because they provide services and the DHCP pool starts at 192.168.10.10, so there is no clash.
- Ping from every PC to the router (192.168.10.1), to the server (192.168.10.2) and between PCs gave 0% loss.
- The name www.mitwpu.edu resolved to 192.168.10.2 through the DNS A record, and the web page opened in the browser over HTTP (port 80).
- User1 sent a mail to user2 through SMTP and user2 downloaded it through POP3, which confirms the email service works.
- When the DHCP service was turned off and a PC was renewed, the PC fell back to an APIPA address (169.254.x.x) and could not reach the server. This shows why DHCP is needed.

FAQ (Handwritten & Scanned)

1. What are the different DHCP messages involved in the dynamic IP address allocation process?

Message	Sent by	Purpose
DHCPDISCOVER	Client (broadcast)	Finds available DHCP servers on the network.
DHCPOFFER	Server	Offers an IP address, mask, gateway, DNS and lease time.
DHCPREQUEST	Client (broadcast)	Accepts one offer (and declines others), or renews a lease.
DHCPACK	Server	Confirms the lease and the configuration.
DHCPNAK	Server	Rejects a request (address invalid or already in use).
DHCPDECLINE	Client	Tells the server the offered address is already in use.
DHCPRELEASE	Client	Gives the address back before the lease ends.
DHCPINFORM	Client	Asks for extra settings when the IP is already set manually.

2. Explain the DORA process used by DHCP for dynamic IP address allocation.

- **Discover:** a new client has no IP, so it broadcasts from 0.0.0.0:68 to 255.255.255.255:67 to find a server.
- **Offer:** each server that has a free address replies with an offered IP, mask, gateway, DNS and lease time.
- **Request:** the client broadcasts a request for one offer, which also tells the other servers to withdraw theirs.
- **Acknowledge:** the chosen server confirms the lease and the client starts using the address. The client tries to renew at 50% of the lease time (T1) and again at 87.5% (T2).

3. What are the different types of Resource Records (RRs) used in DNS? Explain their purpose.

Record	Purpose	Example
A	Maps a host name to an IPv4 address	www.mitwpu.edu A 192.168.10.2
AAAA	Maps a host name to an IPv6 address	www AAAA 2001:db8::2
CNAME	Alias that points to another (canonical) name	web CNAME www.mitwpu.edu
NS	Names the authoritative name server of a zone	mitwpu.edu NS ns1.mitwpu.edu
MX	Names the mail server of a domain, with priority	mitwpu.edu MX 10 mail.mitwpu.edu
PTR	Maps an IP address back to a name (reverse lookup)	2.10.168.192.in-addr.arpa PTR www
SOA	Holds zone details: primary server, serial, refresh timers	one per zone
TXT	Holds free text, used for SPF and domain verification	"v=spf1 mx -all"

4. Compare SMTP, POP3, and IMAP based on their functions and methods of handling email messages.

Point	SMTP	POP3	IMAP
Function	Sends and relays mail	Downloads mail from server	Reads and manages mail on server
Direction	Push (client to server, server to server)	Pull	Pull and sync
Ports	25, 587, 465 (SSL)	110, 995 (SSL)	143, 993 (SSL)
Mail storage	Does not store for the user	Local device; server copy usually deleted	Stays on the server
Multi-device use	Not applicable	Poor, mail sits on one device	Good, all devices see the same mailbox
Folders / flags	No	No (local only)	Yes, kept on server
Offline access	No	Yes, after download	Limited, needs cache